

Problem

Java's `BitSet` class implements a vector of bit values (i.e.: *false* (0) or *true* (1)) that grows as needed, allowing us to easily manipulate bits while optimizing space (when compared to other collections). Any element having a bit value of **1** is called a set bit.

Given **2** BitSets, **B_1** and **B_2** , of size **N** where all bits in both BitSets are initialized to **0**, perform a series of **M** operations. After each operation, print the number of set bits in the respective BitSets as two space-separated integers on a new line.

Submissions

Input Format

The first line contains **2** space-separated integers, **N** (the length of both BitSets **B_1** and **B_2**) and **M** (the number of operations to perform), respectively.

The **M** subsequent lines each contain an operation in one of the following forms:

Leaderboard

- **AND** **<set>** **<set>**
- **OR** **<set>** **<set>**
- **XOR** **<set>** **<set>**
- **FLIP** **<set>** **<index>**
- **SET** **<set>** **<index>**

Discussions

In the list above, **<set>** is the integer **1** or **2**, where **1** denotes **B_1** and **2** denotes **B_2** .

<index> is an integer denoting a bit's index in the BitSet corresponding to **<set>**.

For the binary operations **AND**, **OR**, and **XOR**, operands are read from left to right and the BitSet resulting from the operation replaces the contents of the first operand. For example:

Editorial

AND 2 1

B_2 is the left operand, and **B_1** is the right operand. This operation should assign the result of **$B_2 \wedge B_1$** to **B_2** .

Constraints

- **$1 \leq N \leq 1000$**
- **$1 \leq M \leq 10000$**

Output Format

After each operation, print the respective number of set bits in BitSet **B_1** and BitSet **B_2** as **2** space-separated integers on a new line.

Sample Input

```
5 4
AND 1 2
SET 1 4
FLIP 2 2
OR 2 1
```

Sample Output

Change Theme

Language

Java 15



```
1 import java.io.*;
2 import java.util.*;
3
4 public class Solution {
5
6     public static void main(String[] args) {
7         /* Enter your code here. Read input from STDIN. Your class should
8            Print output to STDOUT. Your class should
9            Solution.*/
10    }
11 }
```

Line: 1 Col: 1

Upload Code as File

Run Code

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☐ Test against custom input